WORKSHEET

Chapter 4: Recurrence relations for arithmetic and geometric sequences

1 Find the recurrence relation for each sequence:

a 2, 6, 10, 14, …

b 19, 16, 13, 10, …

c 2, 4, 8, 16, …

d 2, 6, 18, 54, …

e 7, −2, −11, −20, …

f 5, −10, 20, −40, …

g $2, 1, \frac{1}{2}, \frac{1}{4}, \ldots$

h −4, −2.5, −1, 0.5, …

2 Find the rule for the nth term of the sequence defined by each of the following recurrence relations.

a $t_{n+1} = t_n + 7, t_1 = 4$

b $t_{n+1} = t_n - 6, t_1 = 20$

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c $t_{n+1} = t_n + 3.5, t_1 = 2$

d $t_{n+1} = t_n - 12, t_1 = 40$

e $t_{n+1} = 3t_n, t_1 = 2$

f $t_{n+1} = 2t_n, t_1 = -5$

g $t_{n+1} = \frac{1}{2}t_n, t_1 = 32$

h $t_{n+1} = 5t_n, t_1 = -0.3$

3 The third term of an arithmetic sequence is 8 and the common difference is 3.

a Find the first term of this sequence.

b Write the recurrence relation that describes this sequence.

4 The fifth term of an arithmetic sequence is -12 and the common difference is -4.

a Find the first term of this sequence.

b Write the recurrence relation that describes this sequence.

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5 The fourth term of an arithmetic sequence is 7.2 and the common difference is 2.1.

a Find the first term of this sequence.

b Write the recurrence relation that describes this sequence.

6 The third term of an arithmetic sequence is 15 and the 10th term is 57.

a Find the common difference of this sequence.

b What is the first term of this sequence?

c Find the recurrence relation that defines this sequence.

7 The fifth term of an arithmetic sequence is -8 and the 14th term is -26.

a Find the common difference of this sequence.

b What is the first term of this sequence?

c Find the recurrence relation that defines this sequence.

8 The ninth term of an arithmetic sequence is 48 and the 20th term is 119.5.

a Find the common difference of this sequence.

b What is the first term of this sequence?

c Find the recurrence relation that defines this sequence.

9 The fourth term in a geometric sequence is 40 and the common ratio is 2.

a Find the first term of the sequence.

b Write the recurrence relation that describes this sequence.

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10 The eighth term in a geometric sequence is 1 and the common ratio is 0.5.

a Find the first term of the sequence.

b Write the recurrence relation that describes this sequence.

11 The fifth term in a geometric sequence is 405 and the common ratio is 3.

a Find the first term of the sequence.

b Write the recurrence relation that describes this sequence.

12 The first term of a geometric sequence is -2 and the fourth term is -250.

a Find the common ratio for this sequence.

b Find the recurrence relation of the sequence.

13 The third term of a geometric sequence is 72 and the fifth term is 648.

If $r > 0$:

a find the common ratio for this sequence

b find the recurrence relation of the sequence.

14 The second term of a geometric sequence is -6 and the sixth term is -96.

Given that $r < 0$:

a find the common ratio for this sequence

b find the recurrence relation of the sequence.

15 Each equation below defines a rule for the nth term of a sequence in terms of n. For each one, find the recurrence relation for the nth term of the sequence.

a $t_n = 4 + n$

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b $t_n = 5(3)^{n-1}$

c $t_n = 7(2)^{n-1}$

d $t_n = 13 - 3n$

e $t_n = -3(2)^{n-1}$

f $t_n = 6n - 14$

g $t_n = 37 - 7n$

h $t_n = 270\left(\frac{1}{3}\right)^{n-1}$

16 State whether the sequences that obey the following recurrence relations are arithmetic, geometric or neither.

a $t_{n+1} = 4t_n$ where $t_1 = 5$

b $t_{n+1} = 5t_n + 2$ where $t_1 = 1$

c $t_{n+1} = t_n + 8$ where $t_1 = -17$

d $t_{n+1} = 3 - 4t_n$ where $t_1 = 6$

e $t_{n+1} = t_n - 1.6$ where $t_1 = 12.5$

f $t_{n+1} = 0.3t_n$ where $t_1 = 10$

g $t_{n+1} = -17t_n$ where $t_1 = 285$

h $t_{n+1} = t_n + 1.3$ where $t_1 = -0.5$

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Answers

1 a $t_{n+1} = t_n + 4$ where $t_1 = 2$

b $t_{n+1} = t_n - 3$ where $t_1 = 19$

c $t_{n+1} = 2t_n$ where $t_1 = 2$

d $t_{n+1} = 3t_n$ where $t_1 = 2$

e $t_{n+1} = t_n - 9$ where $t_1 = 7$

f $t_{n+1} = -2t_n$ where $t_1 = 5$

g $t_{n+1} = \frac{1}{2}t_n$ where $t_1 = 2$

h $t_{n+1} = t_n + 1.5$ where $t_1 = -4$

2 a $t_n = 7n - 3$

b $t_n = 26 - 6n$

c $t_n = 3.5n - 1.5$

d $t_n = 52 - 12n$

e $t_n = 2(3)^{n-1}$

f $t_n = -5(2)^{n-1}$

g $t_n = 32\left(\frac{1}{2}\right)^{n-1}$

h $t_n = -0.3(5)^{n-1}$

3 a 2

b $t_{n+1} = t_n + 3$ where $t_1 = 2$

4 a -28

b $t_{n+1} = t_n - 4$ where $t_1 = -28$

5 a 0.9

b $t_{n+1} = t_n + 2.1$ where $t_1 = 0.9$

6 a 6

b 3

c $t_{n+1} = t_n + 6$ where $t_1 = 3$

7 a -2

b 0

c $t_{n+1} = t_n - 2$ where $t_1 = 0$

8 a 6.5

b -4

c $t_{n+1} = t_n + 6.5$ where $t_1 = -4$

9 a 5

b $t_{n+1} = 2t_n$ where $t_1 = 5$

10 a 128

b $t_{n+1} = 0.5t_n$ where $t_1 = 128$

11 a 5

b $t_{n+1} = 3t_n$ where $t_1 = 5$

12 a 5

b $t_{n+1} = 5t_n$ where $t_1 = -2$

13 a 3

b $t_{n+1} = 3t_n$ where $t_1 = 8$

14 a -2

b $t_{n+1} = -2t_n$ where $t_1 = 3$

15 a $t_{n+1} = t_n + 1$ where $t_1 = 5$

b $t_{n+1} = 3t_n$ where $t_1 = 5$

c $t_{n+1} = 2t_n$ where $t_1 = 7$

d $t_{n+1} = t_n + 3$ where $t_1 = 10$

e $t_{n+1} = 2t_n$ where $t_1 = -3$

f $t_{n+1} = t_n + 6$ where $t_1 = -8$

g $t_{n+1} = t_n - 7$ where $t_1 = 30$

h $t_{n+1} = \frac{1}{3}t_n$ where $t_1 = 270$

16 a Geometric

b Neither

c Arithmetic

d Neither

e Arithmetic

f Geometric

g Geometric

h Arithmetic

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